TECHMANTHAN 2026

AI & Smart Cities

AI-Powered Smart City Resource Allocation System

TEAM

ATHARVA

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INSTITUTION

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PROBLEM STATEMENT

What is the Problem?

Cities generates massive real-time data from traffic junctions, waste bins, and emergency zones — but city officials have no unified system to act on It be specific and concise

No predictive capability — decisions are reactive, not proactive

Resource deployment (police, waste trucks, ambulances) is manual and guesswork-based

Who is Affected?

City officials and traffic control authorities (direct users of our system)

Entire urban population of any city — drivers, residents, emergency patients

Why is it Important?

Traffic congestion like 8 Udaipur junctions (Delhi Gate, Surajpol, Hiran Magri, etc.) spikes by 150–400 vehicles during rush hours with no adaptive signal control

Waste overflow in areas like Bhupalpura and Hiran Magri creates public health risks — bin fill can reach 100% with no automatic dispatch trigger-

Emergency incidents spike in rainy weather (+2–6 incidents) and at night (22:00–03:00) but response stays fixed regardless

REAL-WORLD EXAMPLE

[Add a compelling real-world scenario, news report, or statistic that illustrates the severity of this problem]

PROBLEM ANALYSIS

Current Challenges

- Udaipur generates massive real-time data from traffic junctions, waste bins, and emergency zones — but city officials have no unified system to act on it-
- No predictive capability — decisions are reactive, not proactive-
- Resource deployment (police, waste trucks, ambulances) is manual and guesswork-based

Existing Solutions

- Basic CCTV-based traffic monitoring (static, no prediction)-
- Fixed waste collection schedules regardless of bin fill levels-
- Manual emergency dispatch with no risk-based prioritization

Gaps in Current Systems

- No unified platform combining all three domains-
- Systems show data but give no actionable recommendations-
- Zero predictive capability — no system anticipates congestion, overflow, or emergency spikes

PROPOSED SOLUTION

YOUR IDEA

We built a unified AI dashboard for Udaipur city officials that ingests real-time simulated sensor data across traffic, waste, and emergency domains. A FastAPI backend runs ML models every few seconds and serves deployment recommendations to operators via a live Next.js dashboard

How It Solves the Problem

Key Features

- Traffic Prediction: Random Forest model predicts high congestion at any of 8 Udaipur junctions based on hour, day, weather, and vehicle count-
- Waste Overflow Prediction: ML model predicts bin overflow risk based on area, population density, days since last collection, and current fill %-
- Decision Engine: /system/decision API aggregates all three domains and outputs specific actions (e.g., "Deploy traffic police at congested junction")-
- Live City Map: Folium map of Udaipur showing traffic heatmap (red), waste overflow markers (orange), emergency high-risk zones (blue)

INNOVATION & UNIQUENESS

What Makes It Different?

- Udaipur-specific, not generic: All 8 junctions, 8 waste areas, and 6 emergency zones mapped to real GPS coordinates of Udaipur city-
- Three domains, one platform: Traffic + Waste + Emergency unified in a single /system/decision endpoint — no existing Indian city tool does this
- Actionable, not just visual: System doesn't just show data — it tells officials what to do (deploy officers, dispatch trucks, send ambulances)-
- Real-time simulation engine: City state updates every 5 seconds via background async loop, mimicking live sensor behavior

Why Is It Better?

- | Feature | Generic Dashboards | Our System ||---|---|---|| City-specific coordinates | ❌ | ✅ Udaipur GPS mapped || Predictive ML | ❌ | ✅ Random Forest trained || Actionable recommendations | ❌ | ✅ Via Decision Engine || Multi-domain unified | ❌ | ✅ Traffic+Waste+Emergency || Real-time update | ❌ | ✅ Every 5 seconds |

SYSTEM ARCHITECTURE

[INSERT ARCHITECTURE DIAGRAM HERE]

Use draw.io, Lucidchart, or Excalidraw
to create your system flow diagram

Components

- Frontend: [React / Flutter]
- Backend: [Node / Django]
- AI/ML: [Model name]
- Database: [MongoDB / SQL]
- APIs: [list key APIs]
- Cloud: [AWS / GCP / Azure]

This slide directly impacts your Technical Approach score

TECHNOLOGY STACK

Languages

- Python
- JavaScript
- TypeScript

Frameworks

- - Next.js 16.2 (React 19) —
- Dashboard- Tailwind CSS —
- UI Styling- Recharts —
- Real-time trend charts- React-Leaflet + Leaflet —
- Interactive city map- Framer Motion — Animated components

AI/ML Models

- Scikit-learn RandomForestClassifier
- Traffic model (features: hour, day, junction, weather, vehicles → predicts high_congestion)- Scikit-learn RandomForestClassifier
- Waste model (features: area, day, density, days since collection, bin fill % → predicts overflow_risk)
- Joblib — Model persistence (.pkl files)

Backend tools

- FastAPI + Uvicorn
- REST API server
- PyJWT — JWT authentication (admin/viewer roles)
- Joblib — ML model serialization

DATASET & DATA USAGE

Dataset Name	Source	Size	Usage
[Dataset 1]	Kaggle / Govt / Custom	[rows / GB]	[Training / Inference]
[Dataset 2]	[Source]	[rows / GB]	[Purpose]

Data Pipeline & Usage

- Data Collection: Structured datasets for traffic, waste, emergency; real-time data simulated
- Preprocessing: Cleaning, encoding categoricals (location, time, weather), feature selection
- Model Training: Lightweight Scikit-learn models; basic train-test split
- Real-time Usage: Simulation engine → ML models → Decision Engine → /system/decision

IMPLEMENTATION PLAN

DAY 1 — Data & Backend (Completed ✓)*

- Generated 2,800 rows of Udaipur-specific training data ✓
- Preprocessed and label-encoded all three datasets ✓
- Trained Random Forest models for traffic and waste prediction ✓
- Built modular FastAPI backend with 7 endpoints ✓
- Implemented async simulation engine (5-second city state updates) ✓
- Built JWT authentication system (admin/viewer roles) ✓

DAY 2 — Frontend & Integration (Completed ✓)

- Built Next.js dashboard with glassmorphism dark UI ✓
- Integrated real-time API polling every 3 seconds ✓
- Implemented live Leaflet map centered on Udaipur [24.5854, 73.7125] ✓-
- Built Recharts trend visualization for traffic and waste metrics ✓
- Generated Folium city map with 3 data layers ✓
- Built Streamlit dashboard as secondary interface ✓

EXPECTED OUTCOME

We use a hybrid data approach combining structured datasets with simulated real-time inputs to replicate realistic urban conditions.

Dashboard / UI

[Real-time monitoring, analytics, visualizations]

AI Output

[Predictions, classifications, recommendations, NLP results]

Alerts / Reports

[Automated notifications, PDF reports, actionable insights]

IMPACT & SCALABILITY

Quantified Impact:

- - System monitors 8 traffic junctions + 8 waste areas + 6 emergency zones simultaneously in real-time-
- Decision engine reduces human analysis time from minutes to under 1 second per cycle-
- Waste dispatch trigger fires automatically when bin fill > 90%, avoiding overflow before it happens-
- Emergency risk scoring identifies high-risk conditions (score > 6.0) before incidents escalate

*Real-World Application:

- Directly applicable to Udaipur's existing ICCC (Integrated Command and Control Centre) infrastructure
- Extensible to all 100 Smart Cities Mission cities — same architecture, swap city coordinates

Scalability Plan

- Horizontal: Modular FastAPI with independent route modules — add new domains without touching existing code-
- Vertical: RandomForest models retrain on real sensor data once deployed
- Multi-city: Coordinate dictionaries in generate_map.py and map_data.py are the only city-specific config

DEMO PLAN

THIS SLIDE IS CRITICAL FOR YOUR FINAL ROUND EVALUATION

Demo Flow

- Step 1: [Show landing / login page]
- Step 2: [Demonstrate core feature]
- Step 3: [Show AI/ML in action]
- Step 4: [Display results / output]
- Step 5: [Edge case handling]

What Will You Show?

- Live prototype / deployed app
- Working AI model output
- UI screenshots / video backup
- Sample test data ready
- Fallback plan if live demo fails

TEAM DETAILS

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[Member 1]

Team Lead / Full-Stack Dev

?

[Member 2]

AI/ML Engineer

?

[Member 3]

Backend Developer

?

[Member 4]

Frontend / UI Designer

?

[Member 5]

Presenter / Research

Thank You!